DevExpress v26.1 Update — Your Feedback Matters

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How To: Create a File Explorer

Nov 13, 2018 • 5 minutes to read

This example demonstrates how to use the **Breadcrumb Edit** control to create a file navigation bar, similar to the one seen in Microsoft Windows Explorer. Two nodes (*Root* and *Computer*) are created at design-time as persistent nodes, always visible to an end-user. Their `BreadCrumbNode.Persistent` and `BreadCrumbNode.PopulateOnDemand` properties must be set to **true**. The *Root* node stores shortcuts to most important directories (Desktop, Windows, Program Files). The *Computer* node allows end-users to quickly navigate through local disks via related shortcuts. The Nodes tree is dynamically generated at runtime according to the current directories hierarchy. To do so, the `RepositoryItemBreadCrumbEdit.ValidatePath`, `RepositoryItemBreadCrumbEdit.NewNodeAdding` and `RepositoryItemBreadCrumbEdit.QueryChildNodes` events are handled.

View Example

Form1.cs

```
using System;
using System.Collections.Generic;
using System.ComponentModel;
using System.Data;
using System.Drawing;
using System.IO;
using System.Linq;
using System.Text;
using System.Threading.Tasks;
using System.Windows.Forms;
using DevExpress.Utils.Menu;
```

```
using DevExpress.XtraEditors;
using DevExpress.XtraEditors.Repository;

namespace FileNavigator {
    public partial class Form1 : XtraForm {
        public static readonly int MaxEntitiesCount = 80;
        public Form1() {
            InitializeComponent();
            Initialize();
        }

        void myItem_Click(object sender, EventArgs e) {
            throw new NotImplementedException();
        }

        void Properties_QueryPopUp(object sender, CancelEventArgs e) {
            throw new NotImplementedException();
        }


        //Set the Breadcrumb Editor's initial path
        void Initialize() {
            breadCrumbEdit1.Path = Environment.GetFolderPath(Environment.SpecialFolder
            foreach (DriveInfo driveInfo in GetFixedDrives()) {
                breadCrumbEdit1.Properties.History.Add(new BreadCrumbHistoryItem(drive
            }
        }

        private void breadCrumbEdit1_Properties_NewNodeAdding(object sender, DevExpres
            e.Node.PopulateOnDemand = true;
        }

        //Check whether or not the target path exists
        private void breadCrumbEdit1_Properties_ValidatePath(object sender, DevExpress
            if (!Directory.Exists(e.Path)) {
                e.ValidationResult = DevExpress.XtraEditors.BreadCrumbValidatePathResu
                return;
            }
            e.ValidationResult = DevExpress.XtraEditors.BreadCrumbValidatePathResult.C
        }

        private void breadCrumbEdit1_Properties_QueryChildNodes(object sender, DevExpr
            //Add custom shortcuts to the 'Root' node
            if (string.Equals(e.Node.Caption, "Root", StringComparison.Ordinal)) {
                InitBreadCrumbRootNode(e.Node);
                return;
            }
            //Add local discs shortcuts to the 'Root' node
            if (string.Equals(e.Node.Caption, "Computer", StringComparison.Ordinal)) {
                InitBreadCrumbComputerNode(e.Node);
```

```
            return;
        }
        //Populate dynamic nodes
        string dir = e.Node.Path;
        if (!Directory.Exists(dir))
            return;
        string[] subDirs = GetSubFolders(dir);
        for (int i = 0; i < subDirs.Length; i++) {
            e.Node.ChildNodes.Add(CreateNode(subDirs[i]));
        }
    }

    void InitBreadCrumbRootNode(BreadCrumbNode node) {
        node.ChildNodes.Add(new BreadCrumbNode("Desktop", Environment.GetFolderPat
        node.ChildNodes.Add(new BreadCrumbNode("Windows", Environment.GetFolderPat
        node.ChildNodes.Add(new BreadCrumbNode("Program Files", Environment.GetFol
    }
    void InitBreadCrumbComputerNode(BreadCrumbNode node) {
        foreach (DriveInfo driveInfo in GetFixedDrives()) {
            node.ChildNodes.Add(new BreadCrumbNode(driveInfo.Name, driveInfo.RootD
        }
    }

    protected BreadCrumbNode CreateNode(string path) {
        string folderName = new DirectoryInfo(path).Name;
        return new BreadCrumbNode(folderName, folderName, true);
    }

    //Get the local drives list
    public static IEnumerable<DriveInfo> GetFixedDrives() {
        foreach (DriveInfo driveInfo in DriveInfo.GetDrives()) {
            if (driveInfo.DriveType != DriveType.Fixed) continue;
            yield return driveInfo;
        }
    }

    //Get all subfolders contained within the target directory
    public static string[] GetSubFolders(string rootDir) {
        string[] subDirs = GetSubDirs(rootDir);
        if (subDirs == null)
            return new string[0];
        if (subDirs.Length <= MaxEntitiesCount)
            return subDirs;
        string[] res = new string[MaxEntitiesCount];
        Array.Copy(subDirs, res, res.Length);
        return res;
    }

    //Get the names of the subdirectories
    public static string[] GetSubDirs(string dir) {
```

```
            string[] subDirs = null;
            try {
                subDirs = Directory.GetDirectories(dir, "*", SearchOption.TopDirectory
            }
            catch { }
            return subDirs;
        }

        private void breadCrumbEdit1_PathChanged(object sender, BreadCrumbPathChangedE
            label2.Text = breadCrumbEdit1.Path;
        }

        private void breadCrumbEdit1_Properties_PathChanged(object sender, BreadCrumbP

        }
    }
}
```

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